THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. When carbon dioxide is passed through lime water it turns: 1

(a) red **(b)** blue **(c)** milky **(d)** green

2. In the nail experiment, the nail in tap water exposed to air rusts because: 1

(a) both air and water are present **(b)** sunlight is present
(c) only water is present **(d)** only air is present

3. Why does curd become sour on keeping long? 1

(a) water evaporates **(b)** sugar is added
(c) it becomes basic **(d)** more lactic acid forms

4. Pencil lead (graphite) in a circuit tester makes the bulb: 1

(a) break **(b)** glow **(c)** stay off **(d)** melt

5. Why are electric cables made of copper wires coated with PVC? 1

(a) PVC is a conductor **(b)** copper conducts, PVC insulates
(c) copper is an insulator **(d)** both are metals

6. Why do we leave the solution undisturbed to get large crystals? 1

(a) slow cooling gives bigger crystals **(b)** shaking gives bigger crystals
(c) heat gives crystals **(d)** light gives crystals

7. A student finds that China rose turns green with a solution. The solution is likely: 1

(a) vinegar **(b)** curd **(c)** lemon juice **(d)** soap solution

8. A student connects a cell, switch and bulb in a loop. The bulb glows only when: 1

(a) the bulb is covered **(b)** the switch is open
(c) the switch is closed **(d)** the cell is removed

9. Which of these is a non-metal that conducts electricity? 1

(a) chlorine **(b)** phosphorus **(c)** sulphur **(d)** graphite

10. A student dissolves salt in water and then evaporates the water. The salt is recovered. This shows: 1

(a) dissolving is a physical change **(b)** salt disappears
(c) salt is a gas **(d)** dissolving is chemical

11. In $2\,HCl + Ca(OH)_2 \longrightarrow CaCl_2 + 2\,H_2O$, the salt formed is: 1

(a) calcium sulphate **(b)** calcium chloride
(c) calcium carbonate **(d)** calcium nitrate

12. The brightness of a bulb increases if: 1

(a) the wire is cut **(b)** the switch is opened
(c) a cell is removed **(d)** more cells are added in series

13. The gas evolved when dilute sulphuric acid reacts with zinc is: 1

(a) oxygen **(b)** hydrogen **(c)** carbon dioxide **(d)** sulphur dioxide

14. A bridge near the sea rusts faster because: 1

(a) sea is deep **(b)** sea is cold
(c) salt in sea air speeds rusting **(d)** sea is blue

15. Why does a tooth paste contain a base? 1

(a) to sweeten **(b)** to make teeth sour
(c) to colour teeth **(d)** to neutralise mouth acids

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why is iron used for making bridges and not sodium? 2

17. Explain how a galvanised bucket resists rusting. 2

18. Why is vinegar, and not baking soda, applied on a sting that is alkaline in nature (such as that of a wasp)? 2

19. Why is a coiled heating element better than a straight wire in a heater? 2

20. Why does aluminium not rust though it is reactive? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Describe the activity to compare the reactivity of zinc, iron and copper with dilute HCl. 3

22. Describe the observations and conclusions when a few drops of lemon juice are added to a baking soda solution. 3

23. Why should antacid tablets not be taken in large amounts without a doctor's advice? 3

24. Explain why we use thick wires for geysers and thin wires for bulbs. 3

OR

Compare the properties of iron and sulphur. 3

25. Why are sweets packed in airtight containers to prevent chemical changes? 3

OR

Give three examples of acids and bases used in cooking. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the full activity of neutralisation of hydrochloric acid with sodium hydroxide using litmus, with equation. 5

OR

Explain the heating effect of current with a simple experiment using a nichrome wire and cells. 5

27. Explain the formation of oxides of magnesium and sulphur, their tests and equations. 5

OR

Explain the changes that occur in a car's exhaust, the engine, the paint and the iron body, and classify them. 5

END OF QUESTION PAPER **Total: 50 Marks**

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